

MobileTutAR: a Lightweight Augmented Reality Tutorial System using Spatially Situated Human Segmentation Videos

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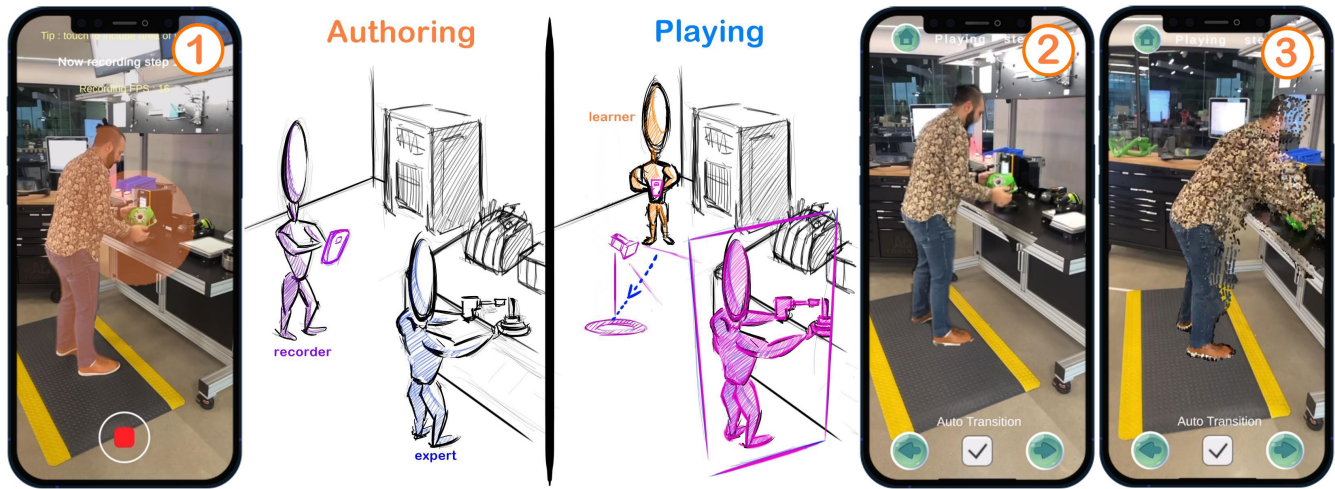


Figure 1: MobileTutAR system workflow. Authoring the tutorial by recording a step-by-step segmentation video of the expert’s demonstration (Left-1). Playing the tutorial by following the navigation guidance and observing the spatially projected 2D images (Right-2) and 3D point cloud (Right-3) human segmentation in AR .

ABSTRACT

Augmented reality (AR) tutorial systems have a strong potential to help workers improve learning efficiency in the ongoing trend of *Industry 4.0*. The current state-of-the-art point cloud approach usually requires cumbersome preparation and additional computational resources only to suffer from low-resolution visual results. To overcome the above limitations, we propose MobileTutAR, a lightweight AR tutorial system that runs entirely on mobile devices providing high definition spatially situated tutorial videos. Our approach captures the tutorial images that extract a human’s body segmentation combined with a user-defined area of interest. The system then projects the tutoring content spatially in situ, to align with the human expert’s recorded position. When playing back, the learner is guided by a navigation-centered user interface to observe the segmentation video from the recorder’s original position/orientation. In this way, we deliver a high-definition AR

experience without needing any cumbersome equipment, exotic computational resources, or in-depth training.

CCS CONCEPTS

• **Information systems** → *Spatial-temporal systems*; • **Human-centered computing** → *User interface toolkits*; *Ubiquitous and mobile computing systems and tools*.

KEYWORDS

tutorial system, spatially situated, augmented reality, human segmentation, machine operation task, lightweight

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1 INTRODUCTION

Intelligent tutorial systems are the key to enabling human workers to master new machine operation tasks in the ongoing trend of *Industry 4.0* [35]. Researchers have been developing new tutoring systems for efficient physical task transfer [24], which have shown a strong potential to reduce training costs and scale up

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distribution. In these novel tutoring systems, Augmented Reality (AR) has been proven to be a particularly effective digital interface [21], due to its nature of seamlessly aligning with the physical reality. This makes AR very useful for machine operation task tutoring when visualizing spatially situated and contextually aware tutorial contents.

The process of creating an AR tutorial for machine operation tasks usually involves recording the demonstration of the human expert while interacting with the physical environment in the work place, and later replaying the captured AR images in situ to the learners. This means the AR tutorial system needs to create a digital copy of the demonstrated task, especially, the human expert's body movement and the object of interaction, with reference to time and space.

The current state-of-the-art approaches for recording AR tutorial tasks are mostly developed around volumetric capture technology, which is enabled by one or more designated depth sensing cameras, such as Microsoft Kinect [5]. This approach uses a 3D point cloud to record and spatially visualize the machine operation task: the human actions and the physical interaction environment. Multiple camera devices can be calibrated and used together to cover a large capture area without too many blind spots [55]. The point cloud can be used to generate a 3D digital model of the captured reality via reconstruction algorithms [38]. Furthermore, this approach can simultaneously achieve human motion capture by extracting 3D skeleton data and then replacing the low-resolution captured human with a virtual avatar for a customized high-definition visualization. The downside of the approach is that setting up the depth sensors in a pre-determined work environment usually requires a lot of time and preparation. The processing and storage of the 3D point cloud can be computationally heavy and therefore requires additional local or cloud-based computing resources. Moreover, the density of the point cloud captured from the room-scale cameras is usually not high enough to preserve important realistic local details, for example: human experts' facial expressions, hand gesture interactions, small machine interface changes, etc.

To overcome the above limitations, our *design goal* is to create an AR tutorial system for machine operation tasks that is capable of supporting: 1) lightweight on-site tutoring workflow in any new environment without installing complex equipment; 2) high resolution AR image data that preserves important human-machine interaction details.

Thanks to the rapid advancement of modern computing technology, mobile devices such as smartphones and tablets are now capable of perceiving the real world through the cameras and other sophisticated sensors better than ever. For example, the latest mobile device products from Apple [7] are equipped with Lidar sensors, powerful microchips, and advanced machine learning algorithms. These enable new AR features such as: robust spatial tracking, true depth sensing, human segmentation, and point cloud volumetric capture [1].

Guided by our design goal and inspired by the state-of-the-art mobile computing technology, we propose MobileTutAR, a lightweight Augmented Reality tutorial system for machine operation tasks using spatially situated human segmentation 2D and volumetric video. Our approach is illustrated in the teaser figure. While capturing the expert's demonstration, we segment out the human

figure and the area of machine interaction from the rest of the camera feed, and record this segmentation image and point cloud data together with the spatial location of the recorder (the one holding the camera) and the recorded (the expert demonstrator), frame by frame (Figure 1 - left - 1). When playing the tutorial to a learner, the segmentation images will be projected onto a canvas that is placed at the recorded expert's spatial location. Meanwhile, the system guides the learner to navigate to the original recorder's location and observe the tutorial content properly (Figure 1 - right). When the learner is observing from the origin recorded position/orientation, the system will play the spatially situated 2D segmentation image, hence producing a high-resolution AR experience (Figure 1 - right - 2). As the learner is moving away from the original recorded location (see the expert's foot matt), the system will seamlessly transit the tutorial content display into 3D point cloud, which preserves the spatiality of the tutoring content in low resolution (Figure 1 - right - 3). To this end, we create an AR tutoring experience with camera resolution details, using standalone mobile devices only. The *contributions* of this paper are described as follows:

- The *workflow* of creating a lightweight AR tutoring experience with spatially situated 2D and 3D human segmentation.
- The *design and implementation* of the MobileTutAR system developed on commercially available mobile devices.

2 RELATED WORK

An extended body of work has focused on improved AR interfaces for tutorial instructions. To start with the related work section, we first specify our criteria of preference: the recorded tutorial task should contain two elements: the human demonstrator and the physical objects of interaction. These related works record such tutorial content while relating it to an AR interface. We summarize the related work into three categories: volumetric 3D capture, human motion capture, and spatially situated 2D video.

2.1 Volumetric 3D capture

Enabled by depth sensing cameras such as Microsoft Kinect [5], point cloud based volumetric capture technology is the state-of-the-art approach for creating holographic effects. With calibration of multiple cameras, this approach is capable of covering a large area including the static object environment, the movable machines and robots, and the human motions as well [8]. This technology is particularly interesting when it's applied for human-centered environmental capture. Past work has explored the idea of time and space manipulation [34] and co-located telepresence [22, 32, 40, 56]. A more noticeable application of the technology is mixed-reality collaboration, where a local user is receiving the AR instruction from a remote virtual reality (VR) user, who is co-located in a scanned/reconstructed virtual environment of the local AR user [12, 30, 42, 44, 48, 50]. More recently, this approach has been adopted for machine operation tasking tutoring and demonstrated for both live tutoring and recorded tutorial in Loki [51]. The limitation of this approach lies mostly with its cumbersome preparations: it requires specific sensing camera devices and heavy computing power to process and store the 3D volumetric data. Also, the point cloud density is usually not good enough for high-resolution reconstruction when captured from a room-scale level.

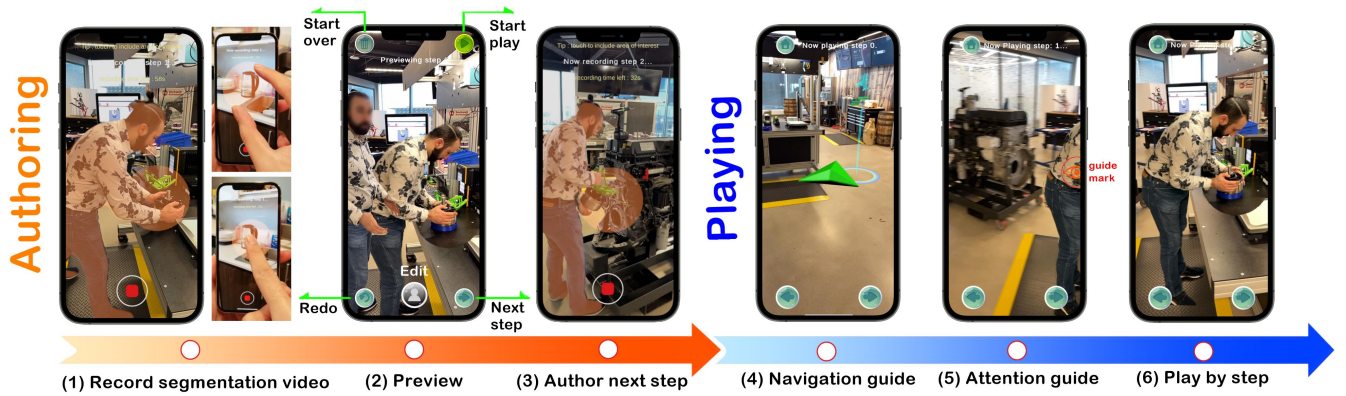


Figure 2: MobileTutAR system design and interaction process. Authoring and playing actions explained step by step.

2.2 Human motion capture

The above mentioned approach can also be used for human motion capture where the system extracts 3D human skeleton poses from images [11]. This technique has been widely applied for body kinetic video games [4]. Similar motion capture results can be achieved by placing markers or additional trackable controllers onto the user, which is widely used in the movie industry [6], as well as the VR gaming field [2, 3]. The method has been broadly adopted in research work to digitize the human motion for avatar-based body movement training [10, 18, 20, 26–28, 33, 37, 39, 49], embodied task authoring [14, 15, 53], and mixed reality remote collaboration [19, 42, 43, 51]. This approach also requires hardware preparation and setup, hence works best in a predefined environment. The skeleton is usually substituted with a virtual avatar and therefore loses the original realistic visual detail of the captured demonstrator. Besides, the motion capture method is usually for capturing human pose only; if additional environmental objects need to be included into captured AR content, then extra prefab modeling and pre-registration work is required.

2.3 Spatially situated 2D video

Video capture is the most popular method of tutorial creation. It's fast, capable of producing high resolution content, and readily available for lightweight devices. It has been broadly adopted into modern tutoring systems providing fine details of operations [16, 17, 25, 31, 52]. Numerous research works have attempted to apply 2D video with AR interfaces for spatially situated tutoring and skill training, including projecting video tutorials onto a nearby physical surface [13, 23, 25, 41, 45–47], or a floating virtual canvas [9, 21, 29, 36, 51, 54]. Despite their popularity, when following the nearby 2D video, users still have to constantly switch attention between the video tutorial and the physical reality of interaction. This leads to distracted attention and has been proven to cause fragmented learning experiences [14].

On the other hand, our approach is an organic combination of traditional 2D video and AR human capture. By segmenting out human action and the object of interaction, we record the core content of the task to project it in situ onto a spatially situated canvas. In

this way, we create a 3D-like AR experience with mostly 2D images, assisted by 3D point cloud. Our method inherits strengths from both sides with a small trade-off of observation perspective constraint, hence delivering a high-definition AR tutoring experience with only lightweight mobile hardware requirements.

3 MOBILETUTAR

3.1 System design and walk-through

In this section, we describe our system design using a walk-through example, which is illustrated in Figure 2. There are two main activities that can be performed with our system. The first activity is authoring, which involves creating the tutorial from an expert demonstration. The second activity is playing, which involves observing the task authored as an AR tutorial in situ.

3.1.1 Authoring. The process of authoring with MobileTutAR is illustrated in Figure 2 (1-3). This process requires at least two people to operate: the recorder who is holding the camera and defining the observation perspective of the AR tutorial, and the recorded expert who is demonstrating the human-machine task. Our system is based on a step-by-step paradigm. Each step should define a subtask that is performed in a specific spatial area. Confining each subtask to a specific area will help obtain a better perspective for the playback from one same point of visualization. The basic pattern of authoring is performed as follows: (1) Author step; (2) Preview step; (3) Confirm and author next step; (4) Repeat until task is completed.

Step recording. We adopt a traditional third person video recording style, a method which most users are already familiar with. While recording, the body of the expert user will be captured in 2D and later displayed in the AR playback, as shown in Figure 2 (1)-left. The core part of the recording is the generation of the body segmentation mask. As seen in the images, this segmentation area is covered by an orange overlay, which gives the authoring user a direct visual feedback when recording the step. The system will automatically include any human figures into the segmentation mask, powered by ARKit's human segmentation feature. Additionally, the authoring user can include an area of interest in order to add the target object of the interaction into the segmentation images. To do this, the user needs to simply touch on the screen with one or

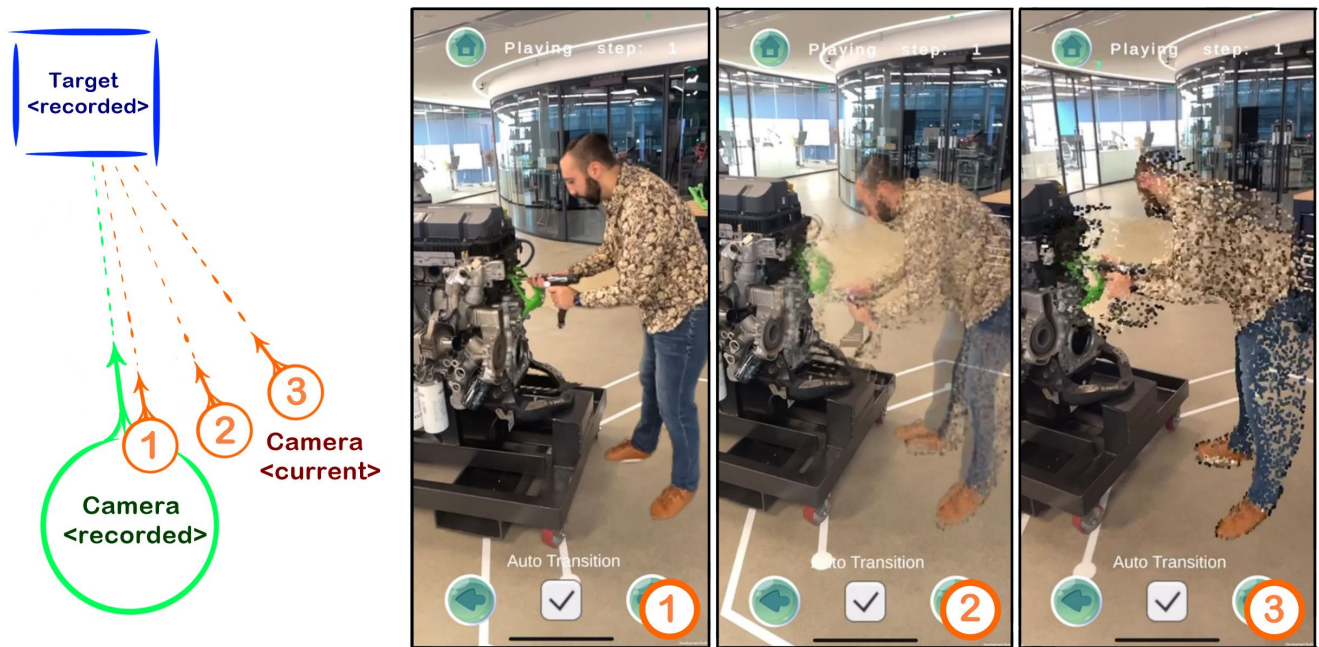


Figure 3: Tutorial visualization with automatic transition between 2D image and 3D point cloud. (1) A High-resolution 2D image will be displayed when the user is currently observing close to the original recorded camera position/orientation; (2) A seamless transition to 3D point cloud will take place when the user is moving away from the recorded position; (3) A Low-resolution 3D point cloud will be displayed when the user is away from the recorded position.

two fingers. Using one finger the user can define the area location with default size and using a second finger, the user can customize the area size, as shown in Figure 2 (1)-right.

Step previewing. When a step recording is finished, the system will take the user to the preview interface, which will instantly play back the current recorded step in loop. At this point, the authoring user can perform one of the following actions: redo the step's recording, cancel the task and start over, move on to author the next step, or finish the recording and start the task playback (Figure 2 (2)). The authoring user can also perform certain edits on the current step, such as changing its visualization style.

3.1.2 Playing. Once the authoring is finished, the tutorial can be played back by a learner, as seen in Figure 2 (4-6). The user is able to visualize in situ in AR the different steps performed by the expert and recorded by the author. Due to the nature of our system, based on a segmented 2D image placed in a 3D canvas, the learner needs to be observing the task from its originally recorded position in order to visualize the AR experience with the appropriate perspective. Our system will guide the user to the observation zone of the step (Figure 2 (4)), and only then will the recorded animation start playing. The system will also guide the user's attention in real-time in case the learner is having a hard time figuring out where to look (Figure 2 (5)). Once in the correct location, the current step animation will play in a loop. The learner can choose to go to next step or to the previous step using the specific UI buttons (Figure 2 (6)).

3D point cloud to 2D image auto-transition To ensure a smooth user experience, we have additionally developed a 3D point cloud feature as an alternative tutorial visualization mode. When the user is away from the original recorded position, instead of showing nothing, we offer an option to show the tutorial content with a low-resolution 3D point cloud. As the user is guided towards the correct observation position/orientation, the tutorial will automatically transit seamlessly to the high-resolution 2D images, as illustrated in Figure 3. It is noted that the point cloud data is recorded together with the 2D image data, and therefore it only contains a 'half-sided' visual content facing towards the camera.

3.2 Implementation

Our prototype system has been developed using Unity3D. The core features that are required for the system are: human segmentation and depth sensing. It is noted that both features are currently available only on iOS mobile devices that support ARKit 3 or higher [1]. The latest iOS devices with Lidar sensor provide more robust depth sensing, but this is not a mandatory requirement since depth can also be estimated with only the RGB camera. In this section of the paper, we will describe the core implementation of the system, which involves: (1) Recording of the human segmentation video; (2) Placement of the segmentation images in 3D location for the AR playback experience; and (3) 3D point cloud visualization on mobile devices.

3.2.1 Human segmentation recording. In order to record the body segmentation, the system captures the following data per frame:

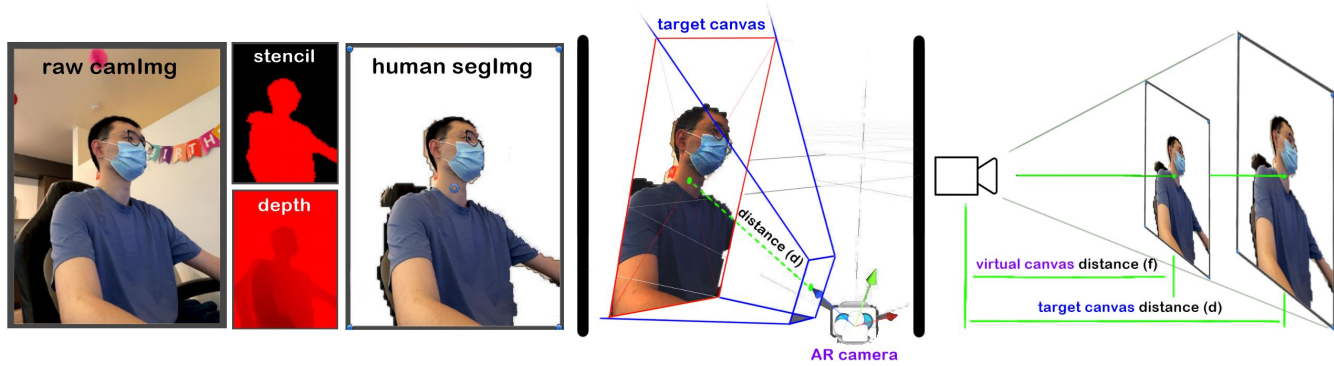


Figure 4: Implementation details of MobileTutAR system. Left: recording human segmentation images. Middle: placing the target canvas with the segmentation images in 3D observation location. Right: scaling the target canvas based on distance.

the texture2D with a transparent channel (RGBA), the position and rotation of the camera, the position and rotation of the target canvas and the distance from the camera to the target canvas (see Figure 4).

The **segmentation texture** is acquired by taking the human stencil (provided by ARKit) as a mask to filter out the body segmentation image from the current camera feed. The result is a RGBA texture where the non-human part has all pixels set to 0 transparency value (Figure 4-Left). A similar method is also used for including a user-defined area of interest into the mask, to include the target object/environment of interaction.

The **position attribute map** is a vector3 array representing the 3D position of the point cloud. It is acquired by reading the raw pixel depth value and transforming it into the world coordinate system.

The **camera position and rotation** are acquired directly by recording the AR camera’s transform information, which is obtained from the spatial tracking enabled by ARKit’s features.

The **target distance** of the segmented human is calculated by taking the average of all the pixel’s depth value that belong to the human stencil mask. Note that the depth API is also provided by ARKit (Figure 4-Left).

The **target position and rotation** are acquired by placing a virtual canvas right in front of the camera. Therefore the target canvas has the same rotation as the camera, and its position is then translated along the camera’s forward vector by the value of the *target distance* (Figure 4-Middle).

3.2.2 AR canvas placement. To display the original recorded segmented images, we need to place the canvas in the proper location in 3D and update its transform. This will be used in the instant preview after authoring each step, as well as in the playback mode. In order to perform the placement, the system will place a canvas (a quad geometry) with the recorded image in the target’s position and rotation at each frame for each step (Figure 4-Middle). Furthermore, the size of the target canvas will need to be scaled properly in real-time according to the distance to match the original AR visual experience. To achieve this, we use an imaginary virtual canvas as a scaling reference that receives the default camera projection.

The size and distance of the virtual canvas are static parameters and can be easily found via calibration experiments. Therefore, the size of the target canvas can be determined by scaling the virtual canvas by d/f , where d is the target canvas distance and f is the virtual canvas distance (Figure 4-Right).

3.2.3 3D point cloud visualization. Point cloud visualization is a very computational-heavy task and it is only possible on mobile devices due to the recent hardware advancement, such as the new microchips that support GPU enabled parallel computing. In our project, we used an iPhone 12 pro, whose CPU image has a native resolution of 2532*1899 pixels. After downsizing to 760*580 pixels for practical purposes, we still have 440800 individual pixels to visualize each frame. In our case, we utilized VFX Graph in Unity3D to facilitate the point cloud visualization as a particle effect. The key inputs for VFX Graph to properly visualize the point cloud are the attribute maps that represent the **color** and **3D position** of each point in the point cloud. The color attribute map is the same as the segmentation texture mentioned previously, and the position attribute map is acquired by reading the raw pixel depth value and transforming it into the world coordinate using Unity3D build-in function: *Camera.ScreenToWorldPoint*. It is worth mentioning that performing this operation 440800 times per frame (once for each pixel) on the CPU during the authoring mode can be very slow (reducing frame-rate to 15-20 FPS) and it is the key factor limiting the performance of the system. Potential improvements include (but not limited to) multi-core CPU based multi-threading, or manually redoing the *Camera.ScreenToWorldPoint* function and use a Compute Shader to load it onto GPU for faster parallel computing.

3.3 Use cases

In order to showcase the system’s application potentials, we present three different examples. In these examples, one expert user is performing three tasks that are recorded by an authoring user. Each one of the use cases exemplifies a specific category of tasks including: assembly, machine operation, and robot manipulation (Figure 5).

Assembly. This first example showcases an expert performing a fan mount assembly for a vehicle engine. The task consists of three



Figure 5: Three example use cases for MobileTutAR: a) Fan mount assembly, b) Machine operation, and c) Robot manipulation.

steps. In the first step, the expert aligns the fan motor into the table fixture. In the second step, the expert locks the mount to the fan motor, and torques the three bolts to complete the sub-assembly. The last step consists of mounting the fan sub-assembly to the truck engine, see Figure 5.

Machine operation. The second example shows an expert operating a feeder machine with a conveyor belt. The task consists of three steps. In the first step, the user empties a bin of lentils into the feeder machine. In the second step, the user places the bin in the scale underneath the conveyor belt. In the last step, the expert activates the conveyor belt by releasing the emergency stop, opening the activation button case, and pressing the button, see Figure 5 (b).

Robot manipulation. The last use case presents an expert manipulating a UR3E robotic arm and an authoring user creating a tutorial, as illustrated in Figure 5 (c). This particular task consists of three steps. In the first step, the expert places a set of items to be picked by the robotic arm onto the conveyor belt. The second step consists of the expert adjusting the position of the arm to the initial configuration. In this step, the expert makes use of his body to interact and configure the position of the robotic arm. The author of the tutorial captures the body of the expert and the part of the robot that is being manipulated. The last step consists of the expert user activating the autonomous motion of the robotic arm. Finally, the playback of the action is being displayed as the tutorial for the task.

4 FUTURE WORK AND CONCLUSION

In this paper, we have introduced a novel approach of delivering lightweight AR tutoring experiences for machine operation

tasks, called MobileTutAR. Our prototype system runs independently on commercially available mobile devices without needing any external sensing devices or cloud computing resources. We have presented the detailed system design and three use cases demonstrating a variety of applications. Future work can focus on improving the system performance by optimizing the image data processing during the recording of the tutorial. A few other related subjects are also interesting to investigate further, such as: How to reduce the misalignment between the 2D situated video and the surroundings? How to duplicate an existing tutorial to apply to other locations? Furthermore, a comprehensive user study is needed to better understand how to balance the visual presentation of 2D segmentation video and 3D point cloud, also their strengths and limitations when compared to the traditional spatially placed unsegmented 2D video. We believe MobileTutAR has demonstrated a novel perspective of combining traditional 2D video recording and modern 3D volumetric capture and has absorbed strengths from both sides in a way that will inspire future researchers.

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